

IIT Hyderabad

Department of Electrical Engineering

Electromagnetics Assignment

Advanced Concepts and Applications

Submitted by:

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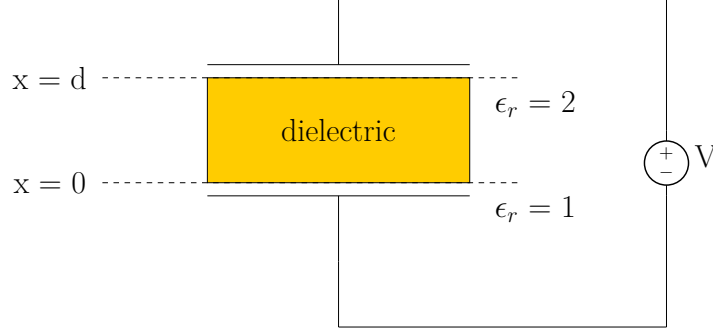
Submitted to:

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Electromagnetics - Semester 2

Q1: The space between the plates of a parallel-plate capacitor is filled with dielectric material whose dielectric constant varies linearly from 1 at the bottom plate ($x = 0$) to 2 at the top plate ($x = d$). The capacitor is connected to a battery of voltage V . Find all the bound charge, and check that the total is zero.



Solution

The diagram above shows the problem including the capacitor and circuit. Since the dielectric constant varies linearly, we can use the boundary values provided to determine the relative permittivity.

$$\epsilon_r(x) = 1 + \frac{x}{d}$$

Symbol	Meaning	Formula
ϵ_0	Permittivity of free space	$8.85 \times 10^{-12} \text{ F/m}$
$\epsilon_r(x)$	Relative permittivity (dielectric constant)	$\epsilon_r(x) = 1 + \frac{x}{d}$
D	Displacement field (independent of dielectric)	$D = \epsilon_0 \epsilon_r(x) E$
$E(x)$	Electric field (varies due to $\epsilon_r(x)$)	$E(x) = \frac{D}{\epsilon_0 \epsilon_r(x)}$
σ_{free}	Free charge density on plates	$\sigma_{\text{free}} = D \text{ (at } x = d)$
σ_b	Surface bound charge density	$\sigma_b = \left(1 - \frac{1}{\epsilon_r}\right) \sigma_{\text{free}}$
$\rho_b(x)$	Volume bound charge density	$\rho_b = -\nabla \cdot (D - \epsilon_0 E)$

Since there is no free charge inside the dielectric, D is constant:

$$D = D \hat{x}$$

$$V = \int_0^d E(x) dx = \frac{D}{\epsilon_0} \int_0^d \frac{dx}{1 + \frac{x}{d}}$$

Let $u = 1 + \frac{x}{d}$, $du = \frac{dx}{d}$:

$$V = \frac{Dd}{\epsilon_0} \int_1^2 \frac{du}{u} = \frac{Dd}{\epsilon_0} \ln 2$$

$$D = \frac{\epsilon_0 V}{d \ln 2}$$

Bound charge appears where ϵ_r changes abruptly (at the surfaces).

At $x = 0$:

$$\epsilon_r(0) = 1 \implies \sigma_b(0) = 0$$

(No bound charge here since the dielectric behaves like vacuum.)

At $x = d$:

$$\sigma_b(d) = \left(1 - \frac{1}{\epsilon_r(d)}\right) \sigma_{\text{free}} = \left(1 - \frac{1}{2}\right) \frac{\epsilon_0 V}{d \ln 2} = \frac{\epsilon_0 V}{2d \ln 2}$$

But bound charge opposes free charge, so:

$$\sigma_b(d) = -\frac{\epsilon_0 V}{2d \ln 2}$$

Now, we know that,

$$\rho_b(x) = -\nabla \cdot (D - \epsilon_0 E)$$

$$\therefore \rho_b(x) = -\epsilon_0 \frac{d}{dx} ((\epsilon_r(x) - 1)E(x))$$

$$\rho_b(x) = -\frac{\epsilon_0 V}{d^2 \ln 2} \frac{1}{(1 + x/d)^2}$$

Surface bound charge:

$$Q_{b,\text{surface}} = \sigma_b(d) \cdot A = -\frac{\epsilon_0 V A}{2d \ln 2}$$

Volume bound charge:

$$Q_{b,\text{volume}} = \int_0^d \rho_b(x) A dx = -\frac{\epsilon_0 V A}{d \ln 2} \int_1^2 \frac{du}{u^2} = -\frac{\epsilon_0 V A}{d \ln 2} \left(-\frac{1}{u} \Big|_1^2 \right) = \frac{\epsilon_0 V A}{2d \ln 2}$$

Total bound charge:

$$Q_{b,\text{total}} = Q_{b,\text{surface}} + Q_{b,\text{volume}} = -\frac{\epsilon_0 V A}{2d \ln 2} + \frac{\epsilon_0 V A}{2d \ln 2} = 0$$

So,

- Surface bound charge at $x = d$: $\sigma_b = -\frac{\epsilon_0 V}{2d \ln 2}$
- Volume bound charge density: $\rho_b(x) = -\frac{\epsilon_0 V}{d^2 \ln 2} \frac{1}{(1 + x/d)^2}$
- Total bound charge is zero, as required

Q2: An electric dipole $\mathbf{p}$, pointing in the y direction, is placed midway between two large conducting plates, as shown in Fig. 1. Each plate makes a small angle θ with respect to the x axis, and they are maintained at potentials $\pm V$. What is the direction of the net force on $\mathbf{p}$? Draw the diagram to justify your answer.

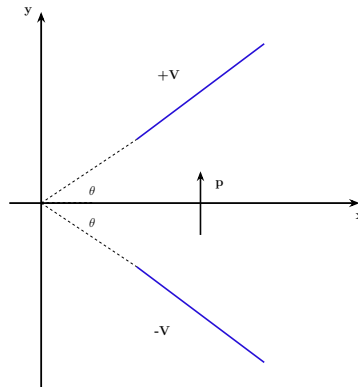


Figure 1:

Solution

- **Dipole Orientation:** The dipole $\mathbf{p}$ points in the y -direction, i.e., $\mathbf{p} = p\hat{y}$.
- **Plate Configuration:** The two conducting plates are inclined at a small angle θ with respect to the x -axis and are maintained at potentials $+V$ and $-V$.
- **Position of Dipole:** The dipole is placed midway between the plates, where the distance between the plates is small compared to their length, creating an approximately uniform electric field in the region near the dipole.

The plates are at potentials $+V$ and $-V$, and the distance between them (along the y -direction) is d . The electric field $\mathbf{E}$ between two parallel plates is:

$$\mathbf{E} = -\frac{2V}{d}\hat{y},$$

where the factor of 2 arises because the dipole is midway between the plates, and the total potential difference is $2V$.

However, since the plates are inclined at an angle θ , the electric field has both x - and y -components. For small θ , the dominant component is still in the y -direction, with a small x -component:

$$\mathbf{E} \approx -\frac{2V}{d}\hat{y} + E_x\hat{x},$$

where E_x arises due to the tilt.

The net force on an electric dipole in an external electric field is given by:

$$\mathbf{F} = (\mathbf{p} \cdot \nabla)\mathbf{E}.$$

Since $\mathbf{p} = p\hat{y}$, this simplifies to:

$$\mathbf{F} = p \frac{\partial \mathbf{E}}{\partial y}.$$

- The y -component of $\mathbf{E}$ ($E_y = -\frac{2V}{d}$) is **constant** between the plates (for small θ), so $\frac{\partial E_y}{\partial y} = 0$.
- The x -component of $\mathbf{E}$ (E_x) **varies with** y because the plates are tilted. As y increases, the distance between the plates changes, altering E_x . Thus:

$$\frac{\partial E_x}{\partial y} \neq 0.$$

For the given tilt, $\frac{\partial E_x}{\partial y}$ is **negative** (as y increases, E_x decreases).

Therefore, the force components are:

$$F_x = p \frac{\partial E_x}{\partial y} \quad (\text{non-zero, negative}),$$

$$F_y = p \frac{\partial E_y}{\partial y} = 0.$$

Therefore, the net force on the dipole is in the **negative x -direction**.

The below diagram justifies the force direction as required.

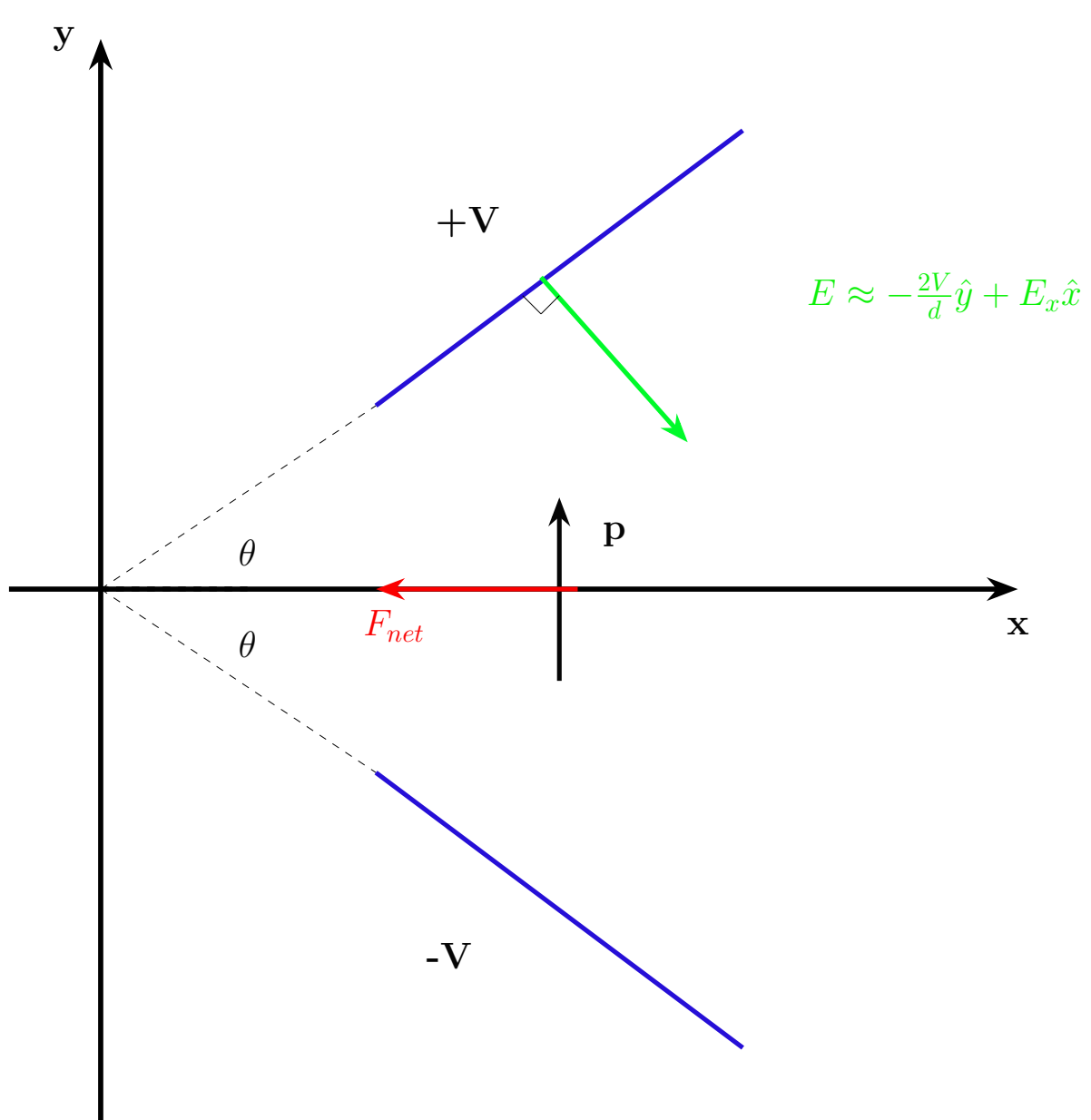


Figure 2:

Q3: The space between the plates of a parallel-plate capacitor is filled with two slabs of linear dielectric material. Each slab has thickness a , so the total distance between the plates is $2a$. Slab 1 has a dielectric constant of 2, and slab 2 has a dielectric constant of 1.5. The free charge density on the top plate is σ and on the bottom plate $-\sigma$. Draw a figure showing the mentioned situation and find the following.

- Find the electric displacement $\mathbf{D}$ in each slab.
- Find the electric field $\mathbf{E}$ in each slab.
- Find the polarization $\mathbf{P}$ in each slab.
- Find the potential difference between the plates.
- Find the location and amount of all bound charge.

f) Now that you know all the charge (free and bound), recalculate the field in each slab, and confirm your answer to (b).

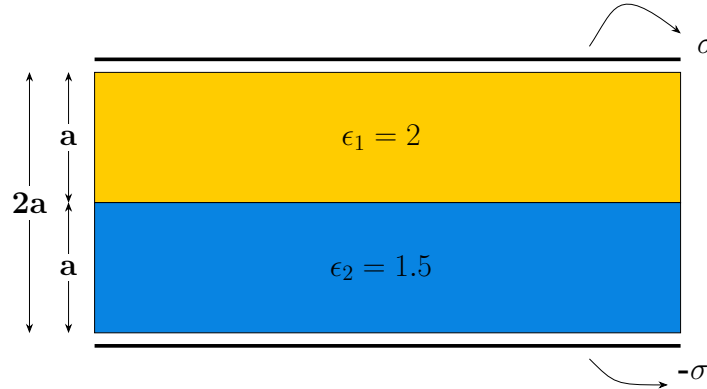


Figure 3: Diagram

Solution

Part (a)

We know that, the displacement vector $\vec{D}$ is related to free charge density only. Now, since the free charge density of both the slabs is same, so $\vec{D}$ is same for both slabs. According to Gauss's law,

$$\oiint_S \vec{D} \cdot d\vec{A} = Q_{\text{free}}$$

Consider a Gaussian pillbox enclosing part of the top plate and extending into the dielectric.

- The free charge enclosed is σA , where A is the area of the plate.
- Since D is uniform, constant and perpendicular to the surface, so the Gauss's law reduces to $DA = \sigma A$

So,

$$\boxed{\vec{D} = \sigma \hat{n}}$$

Part (b)

We know that,

$$\vec{D} = \epsilon_0 \epsilon_r \vec{E}$$

So,

- For slab 1,

$$\sigma \hat{n} = \epsilon_0(2) \vec{E}_1 \implies \boxed{\vec{E}_1 = \frac{\sigma}{2\epsilon_0} \hat{n}}$$

- For slab 2,

$$\sigma \hat{n} = \epsilon_0(1.5) \vec{E}_2 \implies \boxed{\vec{E}_2 = \frac{2\sigma}{3\epsilon_0} \hat{n}}$$

Part (c)

Polarization $\vec{P}$ is related to $\vec{D}$ and $\vec{E}$ by the relation:

$$\vec{P} = \vec{D} - \epsilon_0 \vec{E}.$$

So,

- For slab 1,

$$\vec{P}_1 = \vec{D} - \epsilon_0 \vec{E}_1 \implies \vec{P}_1 = \sigma \hat{n} - \epsilon_0 \left(\frac{\sigma}{2\epsilon_0} \hat{n} \right)$$

$$\boxed{\vec{P}_1 = \frac{\sigma}{2} \hat{n}}$$

- For slab 2,

$$\vec{P}_2 = \vec{D} - \epsilon_0 \vec{E}_2 \implies \vec{P}_2 = \sigma \hat{n} - \epsilon_0 \left(\frac{2\sigma}{3\epsilon_0} \hat{n} \right)$$

$$\boxed{\vec{P}_2 = \frac{\sigma}{3} \hat{n}}$$

Part (d)

The potential difference is the integral of the electric field across the distance between the plates.

Since, the electric field is constant in each slab,

$$\phi = |\vec{E}_1| \cdot a + |\vec{E}_2| \cdot a$$

$$\phi = \frac{\sigma a}{2\epsilon_0} + \frac{2\sigma a}{3\epsilon_0}$$

$$\boxed{\phi = \frac{7\sigma a}{6\epsilon_0}}$$

Part (e)

Bound charges can be volume or surface charges.

- **Volume Bound Charge:**

$$\rho_b = -\nabla \cdot \vec{P}$$

Since $\vec{P}$ is uniform within each slab, $\nabla \cdot \vec{P} = 0 \Rightarrow \boxed{\rho_b = 0}$.

- **Surface Bound Charge:**

$$\sigma_b = \vec{P} \cdot \hat{n}$$

where $\hat{n}$ is the unit normal vector pointing out of the dielectric.

So,

- **Top Surface of Slab 1:** Adjacent to the top plate.
 - $\hat{n}$ points upwards (out of the dielectric).
 - $\sigma_{b1} = \vec{P}_1 \cdot \hat{n} = P_1 \cos(180^\circ) = -P_1 = -\frac{\sigma}{2}$
- **Bottom Surface of Slab 1 / Top Surface of Slab 2:** Interface between the two slabs.

- For Slab 1, $\hat{n}$ points downwards:

$$\sigma_{b1, \text{bottom}} = \vec{P}_1 \cdot \hat{n} = P_1 \cos(0^\circ) = P_1 = \frac{\sigma}{2}$$

- For Slab 2, $\hat{n}$ points upwards:

$$\sigma_{b2, \text{top}} = \vec{P}_2 \cdot \hat{n} = P_2 \cos(180^\circ) = -P_2 = -\frac{\sigma}{3}$$

- At the interface, the net surface bound charge density is:

$$\sigma_{b, \text{interface}} = \sigma_{b1, \text{bottom}} + \sigma_{b2, \text{top}} = \frac{\sigma}{2} - \frac{\sigma}{3} = \frac{\sigma}{6}$$

- By definition (discontinuity of polarization):

$$\sigma_b = \vec{P}_1 \cdot \hat{n} + \vec{P}_2 \cdot \hat{n} = \frac{\sigma}{2} - \frac{\sigma}{3} = \frac{\sigma}{6}$$

- **Bottom Surface of Slab 2:** Adjacent to the bottom plate.

- $\hat{n}$ points downwards.

$$\sigma_{b2, \text{bottom}} = \vec{P}_2 \cdot \hat{n} = P_2 \cos(0^\circ) = P_2 = \frac{\sigma}{3}$$

So, therefore,

- Top surface of Slab 1: $\boxed{-\frac{\sigma}{2}}$
- Interface between Slab 1 and Slab 2 (net effect): $\boxed{\frac{\sigma}{6}}$
- Bottom surface of Slab 2: $\boxed{\frac{\sigma}{3}}$

Part (f)

A parallel-plate capacitor contains two slabs of linear dielectric, each of thickness a , so the total separation is $2a$. The dielectric constants are:

$$\varepsilon_{r1} = 2 \quad (\text{Slab 1}), \quad \varepsilon_{r2} = 1.5 \quad (\text{Slab 2})$$

The free surface charge densities on the plates are:

$$\sigma_{\text{top}} = +\sigma, \quad \sigma_{\text{bottom}} = -\sigma$$

Using parts (a), (b), (c) and (e), we recalculate the electric field using Gauss's law.

Field in Slab 1:

Take a Gaussian pillbox just below the top plate, inside slab 1.

- Enclosed free charge: $+\sigma$
- Bound surface charge at top: $-\frac{\sigma}{2}$

Total charge enclosed:

$$q_{\text{enc}} = \sigma - \frac{\sigma}{2} = \frac{\sigma}{2}$$

$$E_1 = \frac{q_{\text{enc}}}{\varepsilon_0} = \frac{\sigma}{2\varepsilon_0}$$

This matches our earlier result for $\vec{E}_1$.

Field in Slab 2:

Take a Gaussian surface in slab 2 (between interface and bottom plate).

- Bound surface charge at interface (slab 2 side): $-\frac{\sigma}{6}$
- Bound surface charge at bottom: $+\frac{\sigma}{3}$
- Free surface charge at bottom plate: $-\sigma$

Total charge enclosed:

$$q_{\text{enc}} = -\sigma + \frac{\sigma}{3} - \frac{\sigma}{6} = -\frac{2\sigma}{3}$$

$$E_2 = \frac{-2\sigma/3}{\varepsilon_0} = -\frac{2\sigma}{3\varepsilon_0} \Rightarrow \vec{E}_2 = \frac{2\sigma}{3\varepsilon_0} \hat{z}$$

This matches our earlier result for $\vec{E}_2$.

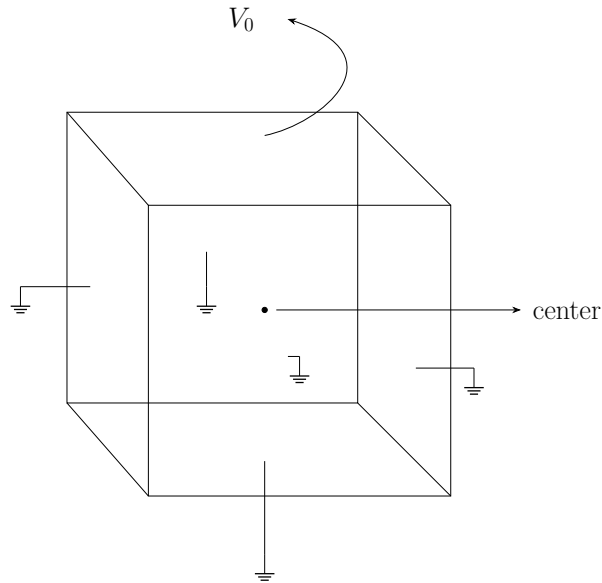
Conclusion:

The recalculated electric fields using all charges (free and bound) are:

$$\vec{E}_1 = \frac{\sigma}{2\varepsilon_0} \hat{z}, \quad \vec{E}_2 = \frac{2\sigma}{3\varepsilon_0} \hat{z}$$

These confirm the values obtained in part (b).

Q4: A cube has 5 sides grounded. The sixth side insulated from the others is held at a potential V_0 . What is the potential at the center of the cube? Why?



Solution

The potential $V(\mathbf{r})$ inside the cube satisfies Laplace's equation with mixed boundary conditions:

$$\nabla^2 V(\mathbf{r}) = 0 \quad \text{for } \mathbf{r} \in \text{interior}$$

$$V(\mathbf{r}) = \begin{cases} 0 & \text{on 5 faces} \\ V_0 & \text{on the insulated face} \end{cases} \quad (1)$$

To solve this boundary value problem:

1. **Uniqueness Theorem:** The solution is unique given the boundary conditions.

2. **Symmetry Argument:**

- The problem has octahedral symmetry about the center when considering image charges
- The potential at the center must be invariant under all symmetry operations of the cube
- The only such potential is the average of all boundary values

3. **Method of Images:**

- We can replace the boundary conditions with an infinite lattice of image cubes
- Each image cube alternates between $+V_0$ and $-V_0$ on corresponding faces
- At the center, contributions from distant images cancel by symmetry
- The dominant term comes from the nearest neighbors, giving $V_0/6$

0.1 Conclusion

The exact potential at the center is:

$$V_{\text{center}} = \frac{V_0}{6}$$

0.2 WHY ?

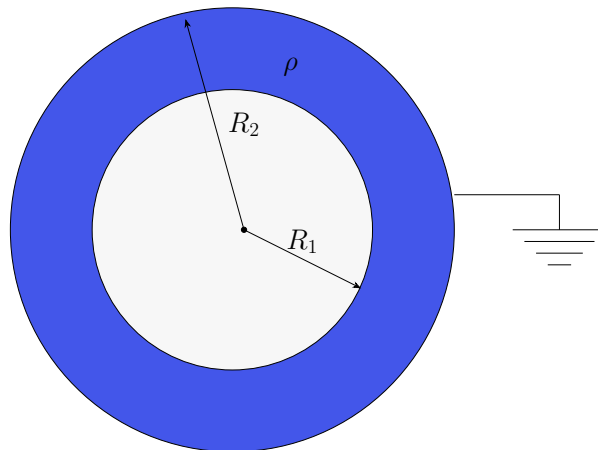
Laplace's Equation Requirement: Inside the charge-free cube, the potential satisfies $\nabla^2 V = 0$. This implies:

- The potential cannot have local maxima or minima (maximum principle)
- The potential must smoothly interpolate between boundary values
- The value at any point is the average of surrounding values

Symmetry Considerations: For the cubic geometry:

- The center is equidistant from all six faces
- The five grounded faces are physically equivalent
- The configuration is symmetric under permutation of the grounded faces

Q5: The inside of a grounded spherical metal shell (inner radius R_1 and outer radius R_2) is filled with space charge of uniform charge density ρ . Find the electrostatic energy of the system? Find the potential at the center?



Solution

Firstly, we determine the electric field $\vec{E}$ in the regions using Gauss's Law.

For $r < R_1$:

$$\vec{E} = 0 \quad (\text{No enclosed charge})$$

For $R_1 \leq r \leq R_2$: Using Gauss's law:

$$\oint \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$E \cdot 4\pi r^2 = \frac{\rho \cdot \frac{4}{3}\pi(r^3 - R_1^3)}{\epsilon_0}$$

$$\vec{E} = \frac{\rho(r^3 - R_1^3)}{3\epsilon_0 r^2} \hat{r}$$

For $r > R_2$:

$$\vec{E} = 0 \quad (\text{Grounded shell shields external field})$$

Now, we proceed for determining the electrostatic energy of the system. The energy density in $R_1 \leq r \leq R_2$:

$$u = \frac{1}{2} \epsilon_0 E^2 = \frac{\rho^2 (r^3 - R_1^3)^2}{18 \epsilon_0 r^4}$$

Total energy:

$$\begin{aligned} U &= \int_{R_1}^{R_2} \frac{\rho^2 (r^3 - R_1^3)^2}{18 \epsilon_0 r^4} \cdot 4\pi r^2 dr \\ &= \frac{2\pi \rho^2}{9 \epsilon_0} \int_{R_1}^{R_2} \left(r^4 - 2R_1^3 r + \frac{R_1^6}{r^2} \right) dr \\ U &= \boxed{\frac{2\pi \rho^2}{9 \epsilon_0} \left(\frac{R_2^5 - R_1^5}{5} - R_1^3 (R_2^2 - R_1^2) - R_1^6 \left(\frac{1}{R_2} - \frac{1}{R_1} \right) \right)} \end{aligned}$$

Finally, we determine the electric potential at the center. Since $\vec{E} = 0$ for $r < R_1$, $V(0) = V(R_1)$. We calculate $V(R_1)$ by integrating from R_2 (where $V = 0$) to R_1 :

$$\begin{aligned} V(R_1) &= - \int_{R_2}^{R_1} \frac{\rho (r^3 - R_1^3)}{3 \epsilon_0 r^2} dr \\ &= - \frac{\rho}{3 \epsilon_0} \int_{R_1}^{R_2} \left(r - \frac{R_1^3}{r^2} \right) dr \\ &= - \frac{\rho}{3 \epsilon_0} \left[\frac{r^2}{2} + \frac{R_1^3}{r} \right]_{R_1}^{R_2} \\ V(0) &= \boxed{- \frac{\rho}{3 \epsilon_0} \left(\frac{R_2^2}{2} + \frac{R_1^3}{R_2} - \frac{3R_1^2}{2} \right)} \end{aligned}$$

So,

- **Electrostatic energy of the system:**

$$U = \boxed{\frac{2\pi \rho^2}{9 \epsilon_0} \left(\frac{R_2^5 - R_1^5}{5} - R_1^3 (R_2^2 - R_1^2) - R_1^6 \left(\frac{1}{R_2} - \frac{1}{R_1} \right) \right)}$$

- **Potential at the center:**

$$V(0) = \boxed{- \frac{\rho}{3 \epsilon_0} \left(\frac{R_2^2}{2} + \frac{R_1^3}{R_2} - \frac{3R_1^2}{2} \right)}$$

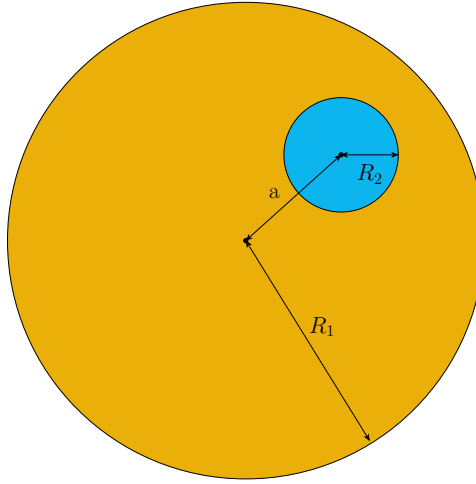
Q6: A sphere of radius R_1 has charge density ρ uniform within its volume, except for a small spherical hollow region of radius R_2 located at distance a from the center.

- Find the field $\vec{E}$ at the center of the hollow sphere.
- Find the potential ϕ at the same point.

Solution

The diagram above clearly illustrates the situation.

Now, for a solid sphere of radius R with uniform charge density ρ :



- **Electric Field** at distance r from center:

$$\vec{E}(r) = \begin{cases} \frac{\rho r}{3\epsilon_0} \hat{r} & (\text{inside, } r \leq R) \\ \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} & (\text{outside, } r > R) \end{cases}$$

where $Q = \frac{4}{3}\pi R^3 \rho$.

- **Potential** at distance r from center (assuming $\phi(\infty) = 0$):

$$\phi(r) = \begin{cases} \frac{\rho}{2\epsilon_0} \left(R^2 - \frac{r^2}{3} \right) & (\text{inside}) \\ \frac{Q}{4\pi\epsilon_0 r} & (\text{outside}) \end{cases}$$

a) Electric Field at the Center of the Hollow Sphere

(a) **Superposition Principle:**

The system can be decomposed into:

- A solid sphere of radius R_1 with uniform charge density ρ .
- A smaller sphere of radius R_2 with charge density $-\rho$ (representing the hollow), centered at displacement $\vec{a}$.

(b) **Electric Field Due to the Solid Sphere:**

Using Gauss's law, the field inside a uniformly charged sphere at distance a is:

$$\vec{E}_{\text{solid}} = \frac{\rho}{3\epsilon_0} \vec{a}.$$

(c) **Electric Field Due to the Hollow Sphere:**

The field at the center of a uniformly charged spherical shell is zero:

$$\vec{E}_{\text{hollow}} = \vec{0}.$$

(d) **Total Electric Field:**

By superposition:

$$\vec{E} = \vec{E}_{\text{solid}} + \vec{E}_{\text{hollow}} = \frac{\rho}{3\epsilon_0} \vec{a}.$$

$$\boxed{\vec{E} = \frac{\rho}{3\epsilon_0} \vec{a}}$$

b) The potential ϕ at the same point

(a) **Potential Due to the Solid Sphere:**

The potential inside a uniformly charged sphere at distance a is:

$$\phi_{\text{solid}} = \frac{\rho}{2\epsilon_0} \left(R_1^2 - \frac{a^2}{3} \right).$$

(b) **Potential Due to the Hollow Sphere:**

The potential at the center of a uniformly charged (negative density) sphere is:

$$\phi_{\text{hollow}} = \frac{-\rho R_2^2}{2\epsilon_0}.$$

(c) **Total Potential:**

By superposition:

$$\phi = \phi_{\text{solid}} + \phi_{\text{hollow}} = \frac{\rho}{2\epsilon_0} \left(R_1^2 - \frac{a^2}{3} - R_2^2 \right).$$

$$\boxed{\phi = \frac{\rho}{2\epsilon_0} \left(R_1^2 - \frac{a^2}{3} - R_2^2 \right)}$$